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Teacher Coaching to Improve Instruction at Scale: Opportunities and Challenges in Policy Contexts

David Blazar
University of Maryland, College Park

Abstract

Background/Context: Teacher coaching, defined by sustained observation and feedback cycles, has a longstanding theory of action that has held up to several rigorous evaluations in recent years. Policymakers and practitioners have taken note of these findings, rolling out coaching programs in classrooms and schools across the United States as a key policy initiative to improve instructional quality and student outcomes at scale. One likely factor leading to its success is that coaching program activities are situated much more closely to key outcomes than other initiatives (e.g., certification, licensure). However, coaching too exists in a broader policy context that creates both opportunities and challenges for success. On one hand, coaches are uniquely positioned not only to help teachers improve their practice but also to serve in leadership and capacity-building roles for their broader education systems; on the other hand, coaches' role in system infrastructures can mean that their work is impeded by these same organizational factors, including top-down hierarchies and competing reform interests.
Purpose/Objective/Research Design: This piece synthesizes qualitative and mixed methods research presented in this volume in order to situate coaching programs within broader and identify avenues for continued exploration.
Conclusions/Recommendations: Navigating these tensions requires additional research and data on the conditions and mechanisms that drive effective teacher coaching, and how these conditions and mechanisms can be replicated at scale. One mechanism that should serve as the focal point of ongoing work is the coach her- or himself. What are the skills and dispositions needed for coaches to work with teachers to develop instructional practice while also effectively navigating policy conditions in their schools and districts?

Instructional Policy *is* Good Education Policy

Improving student outcomes at scale requires wrestling with *the* key input of U.S. educational systems: teachers and the instruction they deliver in class every day. Teachers' salaries and benefits comprise the majority of education expenditures (U.S. Census Bureau, 2016). More importantly, teachers and their instructional practices produce the largest gains in desired student outcomes (Blazar & Kraft, 2017; Hanushek & Rivkin, 2010) relative to investments in other resources including extended learning time, school choice, and student incentives (Fryer, 2017).

Yet, the intensely complex nature of instruction (Cohen, Raudenbush, & Ball, 2003) makes widespread improvements to teacher practice a particularly challenging and, to date, elusive goal. Most interventions aimed at improving teacher and teaching quality -- including teacher certification processes (Boyd, Goldhaber, Lankford, & Wyckoff, 2007), generalized workshops and related professional development (PD) opportunities (Garet et al., 2010; Yoon et al., 2007), and merit pay (Murnane & Cohen, 1986; Podgursky & Springer, 2007) -- have produced disappointing results.

A shining star in the midst of decades worth of weak or null findings is teacher coaching, a type of PD defined by several critical features: coaching programs are individualized, time-intensive, sustained over the course of a semester or year, context-specific, and focused on discrete skills (Joyce & Showers, 1982; Showers, 1985). A recent meta-analysis of 60 studies able to support causal inferences indicate that, on average, coaching programs that share these critical features improve teachers' instructional practice by 0.49 standard deviations (SD) and students' academic performance on standardized tests by 0.18 SD (Kraft, Blazar, & Hogan, 2018). In other words, with coaching, the quality of teachers' instruction improves by as much as -- or more than -- the difference in effectiveness between a novice and a teacher with five to 10 years of experience. In a second meta-analysis of the causal evidence on a range of educational interventions, Fryer (2017) estimates that only high-dosage tutoring produces larger effects on student test scores (0.31 SD); but those effects almost certainly come at a much higher cost than teacher coaching.

Teacher Coaching in a Broader Policy Context

Likely key to the success of coaching programs relative to other teacher-oriented policies is the fact that the intervention is quite proximal to the specific behaviors that it aims to improve: teaching practice and, in turn, student achievement. Coaches work with teachers to observe their classroom behaviors, provide targeted feedback on what they observe, and create a space for teachers to practice new skills.

Comparatively, many other policies aimed at improving teacher and teaching quality (e.g., certification, merit pay) purposefully are much farther removed from the classroom. This distance ensures that the policy can be rolled out broadly, but assumes a trickle-down effect whereby state- or district-level policies are translated effectively into instructional policy in the classroom. The logic models linking these latter teacher-oriented policies to improved instruction and student achievement arguably are much more convoluted than the logic model linking teacher coaching to desired outcomes. (For a visual illustration of a logic model linking teacher

coaching to teacher and student outcomes by myself and colleagues, see Kraft et al., 2018.) And so, evidence of efficacy for coaching but not for other teacher-focused interventions leads to an intuitive conclusion: any effort to successfully improve student outcomes at scale logically must focus directly on helping teachers deliver high-quality instruction (Cohen et al., 2003; Hill, Blazar, & Lynch, 2015; Murnane & Cohen, 1986; Smith, Desimone, & Ueno, 2005).

However, as the authors of this volume effectively point out through a range of analyses on program implementation, coaching too exists in a much broader policy context that creates both opportunities and challenges for success.

Hannan and Russell argue that, given their inherent proximity to classroom instruction, coaches are uniquely positioned not only to help teachers improve their practice but also to serve in leadership and capacity-building roles for their broader education systems. Their case study design and qualitative comparative analyses allow the authors to tease out a combination of conditions that are associated with a given outcome. Applied to a large-scale coaching program implemented in districts across Tennessee, these analyses indicate that coaches often work in conjunction with other school and district leaders who are much more removed from the classroom in order to consider broad efforts to improve instruction at scale.

Galey extends this argument with findings from two additional districts -- one urban and one suburban -- in Indiana using social network analysis. Coaches listed colleagues with whom they discussed coaching and/or support strategies and the frequency of their interactions, resulting in visual diagrams of coaching teams. Analysis of these data revealed that coaches can promote learning not only within a single classroom but also across schools within a district (or broader policy environment). For example, within the specific program under study, coaches leveraged district-wide intranets to build databases of instructional knowledge, thereby enhancing their districts' reform efforts.

At the same time, the conditions of educational systems create unique conditions for coaches and their work. For example, in recent years in Los Angeles Unified School District, principal supervisors took on the role of coach to principals, in addition to overseeing other human capital reforms including teacher evaluation. Thus, the success of coaching as an intervention is inextricably tied to additional factors including constraints on principal time, the position of coaching within broader reform efforts, and dual-role of coaches as evaluator and support provider. Through surveys and interviews with principals and their supervisors in this district, Bush-Mecenas, Marsh, and Strunk examined how principal supervisors coached principals (in terms of their coaching strategies and style) and how this went hand-in-hand with broader support for and high-quality implementation of other human capital reforms.
Unsurprisingly, this and other policy contexts often create challenges for effective implementation of teacher coaching. In Tennessee, Hannan and Russell found that coaches working in larger districts with more teachers to observe required additional structures to feel supported and to engage effectively in their work both with teachers and with school and district leaders.

Similar findings are echoed in Hashim's social network study of a second (anonymous) district in California. Hashim found that coaches' work with teachers around instruction can involve

supporting secondary goals such as building district-wide infrastructures for improvement. Specifically, at the time of broader implementation of the Common Core State Standards with education technology (e.g., iPads and apps), coaches were required to coordinate instructional change efforts with several other "network actors" developing resources for Common Core implementation. However, various organizational factors impeded this work, including the top-down hierarchy of the system-level office, limited school principal support for technology-enabled instruction for Common Core, as well as variation across schools in the interest and ability of teachers to engage with technology necessary to implement coaching.

Finally, by applying concepts of institutionalization, Woulfin provides a useful framing device for coaches' work within broader educational systems and policy contexts. Drawing on interviews, observations, and document analysis from two charter management organizations and one school district, she identifies three factors that shape the degree to which coaching programs can achieve their goal: (1) *organizational conditions*, such as the history of coaching within the system's reform agenda, and the power that teachers' unions wield over out-of-classroom time; (2) *actors* other than coaches in schools and at the district level, who could help facilitate coaching processes or create blockages; and (3) the presence of *established protocols and procedures* (i.e., "artifacts") to guide observation and feedback cycles.

The studies in this volume use qualitative and mixed methods to analyze data from several districts and states across the United States, painting a coherent but complex picture of coaching as implemented in practice. Impact evaluation studies that quantitatively assess the effect of coaching on teacher and student outcomes tell a similar story. In a meta-analysis of the causal evidence on teacher coaching, colleagues and I disaggregated treatment effects for small-scale coaching programs implemented under best-case scenarios versus larger-scale programs implemented in a way that likely is more realistic in a broader policy context (Kraft et al., 2018). We found that, on average, both sets of programs produced positive effects on teachers' practice and students' test-score performance. But, effects were roughly half as large for the latter, larger-scale programs than the former, smaller-scale ones. These findings also are consistent with a large literature on the significant challenges of taking education programs and reform initiatives to scale (Honig, 2006). Continuing to examine and unpack the critical barriers to scale-up and ways to surpass them is critical to the long-term success of coaching as a policy intervention.

Lessons for Future Research

Paired with the theoretical and quantitative research on teaching coaching to date, the rich studies presented in this volume reveal inherent tensions for the success of coaching programs: coaches *are* the intervention with a unique ability to directly influence teachers' practice and other actors with whom they interact as part of their work, yet work in much larger organizational and policy systems with competing interests. Additionally, coaches may have targeted goals of improving an *individual* teacher's classroom practice yet are embedded in systems that are oriented around improving teaching at scale.

Such tensions raise questions that future research can and should examine. Broadly, future research should consider the following: *What are the conditions and mechanisms that drive effective teacher coaching, and how can these conditions and mechanisms be replicated at*

scale? Although we know from the quantitative impact evaluation literature that, on average, coaching improves teachers' instructional practice and student achievement (Kraft et al., 2018), studies of program implementation reveal several factors driving variability across programs. We know much less about how best to address these implementation and policy challenges.

In turn, I suggest that ongoing and future research on teacher coaching must be designed specifically to tease out mechanisms and scalability, which I recognize is not an easy task. Randomized control trials are the gold standard of evaluation research (Murnane & Willett, 2010), and were the primary design of coaching studies included in colleagues and my meta-analysis (Kraft et al., 2018). However, unpacking mechanisms and scalability generally is a challenge in impact evaluation research. For one, the program implementation features that serve as mechanisms and pathways for success by definition are endogenous; they are determined *by* participants rather than by chance. Some mechanisms of coaching programs, such as the number of sessions received or whether coaching is delivered online versus face-to-face are manipulable (for examples, see Powell et al., 2010); one could randomize the number of sessions or mode of coaching, and test the impact of these conditions on changes in teacher practice. However, the mechanisms and conditions described by researchers in this volume that likely are much more critical to the success of coaching programs, including working relationships between coaches and other school or district staff, are much less manipulable by researchers. In addition, while well-designed randomized control trials have strong internal validity, they can lack strong external validity; they cannot provide information on the extent to which a given programs' causal effects will replicate to other populations.

Comparatively, close qualitative analysis of program implementation such as the work presented in this volume help describe the competing factors that may influence the success (or lack thereof) of a specific program. But, they cannot determine definitely whether or not these conditions *caused* the outcomes observed, nor can they generalize beyond a specific program.

In turn, I propose that future research aimed at understanding the mechanisms and scalability of effective teacher must be mixed-methods in nature, whereby specific mechanisms influencing scalability are identified through close qualitative analysis and then tested for efficacy through quantitative impact evaluation designs. I highlight with an example, taking one of the three dimensions for success described by Woulfin (this volume): *actors.* The key actor to effective coaching programs are the corps of coaches hired in a given program. Coaches serve as the primary mechanism for delivering instruction to teachers, and also serve as a key liaison with school and district leadership. In addition to the qualitative analyses in this volume, descriptive quantitative analyses suggest that coaches can vary in their success for carrying out a given program's mission. Examining changes in program effectiveness across two cohorts of a randomized experiment, one colleague and I found that large differences in outcomes likely were attributable, in part, to turnover of coaches; some coaches were associated with very large changes in teacher practice, while other coaches were associated with much smaller or no changes in outcomes (Blazar & Kraft, 2015).

Yet, to date, we know very little about the specific characteristics of effective coaches that is essential for recruiting, training, and supporting coaches to engage in high-quality work with teachers and within broader educational systems. Intuition, theory (Joyce & Showers, 1982;

Showers, 1985), and some empirical evidence (Blazar & Kraft, 2015) highlight likely dimensions of coach quality: (1) professional skills coaches bring to their work (e.g., content knowledge, teaching and leadership experience); (2) tools they use in their interactions with teachers (e.g., providing direct feedback, modeling teaching behavior, opportunities for active practice); and (3) relationships and rapport with teachers and other school and district staff. These dimensions can and should be examined in future research. For example, additional qualitative analyses can provide information to develop instruments and measurements tools to capture variation in these (and other) possible dimensions of coach quality. Random assignment of coaches to teachers would enable researchers to determine definitively the degree to which coaches vary in their ability to improve teachers' instruction and other desired outcomes, and then to examine which dimensions of coach quality are associated with greater versus smaller improvements.

Similar studies that pair qualitative and quantitative research to (a) identify likely mechanisms for scalability, (b) codify these mechanisms in research instruments and tools, and (c) assess the relationship between these dimensions of quality and desired outcomes can and should be carried out to examine several other likely mechanisms of effective coaching programs: school and district climate; presence of competing reform initiatives that create (in)coherent agenda, as well as financial and human capital constraints; degree of standardization and training of coaching procedures; etc.

Concluding Thoughts

In my opinion, teacher coaching is as close to a "silver bullet" as we every will come in education research and policy making. But teacher coaching will not be a silver bullet so long as unique policy contexts inevitably lead to variation in the implementation of these programs. Future research in this area must be designed to help equip districts and schools to effectively navigate policy conditions, and in particular to hire and support educational leaders who are able to implement instructional programs that hold up to external pressures.

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